

Marc Riera CITP

Infrastructure and Platform Engineering Leader | HPC, Storage and Platform Operations | CITP MBCS
Cambridgeshire, UK | marc@riera.co.uk
linkedin.com/in/joanmarcriera/

Profile

Infrastructure and platform engineering leader, formerly Deputy Head of Operations at EMBL-EBI from January 2021 to March 2026, at the institute since 2016. Eight direct reports, about 20 engineers influenced and an annual infrastructure budget of about £1m. Accountable for reliability, cost and delivery of archive, transfer, identity and monitoring services in an estate of more than 4,500 Linux servers and VMs, with more than 150PB of active research data on object storage and tape. Earlier, five years at Bull and Atos installing and operating high-performance computing clusters with Slurm, InfiniBand and Lustre, then managed services and solution design. Infrastructure roles since 2007.

Key results

- FIRE active archive grew from about 15PB at ownership handover to more than 150PB active by March 2026, with about 400PB of raw storage underneath.
- Transfer services: 40 mount points reduced to 3 and 80 virtualised Globus installations to 3. Tickets fell from close to 100 a week to a handful a month; effective throughput rose from 10 to 16Gbps to a sustained 60 to 75Gbps.
- Tape DR: I wrote the tender and secured about £250K; the PoINT Archival Gateway cut disaster recovery total cost of ownership by 70% and RTO from months to days.
- EMBL-EBI's first zero-downtime data-centre migration; I advocated the stretched VLANs behind it.
- Identity for 5,000+ federated users across six EMBL sites; privileged administrators down from about 50 to 3, through a bastion host.

Core skills and technologies

HPC and compute: Slurm, LSF and Sun Grid Engine schedulers; bullx suite; NVIDIA GPU nodes and CUDA drivers on BSC MinoTauro at Bull and Atos, 2011 to 2016; Linpack and IOzone acceptance testing; bare-metal provisioning with Katello, PXE and SaltStack.

Storage and transfer: petabyte object storage and tape; S3 APIs; ActiveScale, Dell EMC Isilon and ECS, NetApp, VAST Data, Lustre on DDN, Ceph; PoINT Archival Gateway on IBM tape; Globus and Aspera.

Networking and data centre: InfiniBand fabric design with Voltaire and Mellanox subnet management; Ethernet bonding and LACP; stretched VLANs; 100Gbps JANET/Jisc link; MetalLB in BGP mode; rack layout, cabling, power and cooling planning, commissioning.

Platform, cloud and automation: Kubernetes on bare metal and virtual machines with kubeadm and RKE; OpenStack; VMware in pre-sales design; AWS and GCP cost modelling; configuration management with Puppet, Ansible and SaltStack; Terraform for virtual load balancers; Rundeck; GitLab CI/CD, mandatory code review, unit and functional tests.

Identity and Access Management: Red Hat Identity Management, FreeIPA, OpenLDAP, Active Directory trust; two-factor onboarding; bastion-host privileged access; joiner and leaver processes; HashiCorp Vault.

Service management and commercial: ITIL 4 incident, request and change practice; change advisory group; Service Desk tiering; ServiceNow; Checkmk, Zabbix, Ganglia; Prometheus and Grafana for storage capacity reporting; availability and disaster recovery planning; tender writing, staged acceptance tests, vendor and supplier management, TCO and cost models.

Relevance to AI infrastructure and GPU cloud

- GPU cluster installation and maintenance at Bull and Atos, 2011 to 2016: BSC MinoTauro with NVIDIA Tesla M2090 GPUs, CUDA drivers, InfiniBand and water-cooled rear doors.
- My production GPU cluster work dates from those years. Since then GPU work has been study and labs: a home lab on TrueNAS with Prometheus and Grafana GPU telemetry on a shared NVIDIA RTX 4060, alongside Ollama and Whisper inference containers, and a multi-GPU, multi-node training and benchmark lab on rented GPU cloud nodes, scheduled for October 2026.
- Scheduler, storage and network operations since then: at EMBL-EBI I helped run three LSF farms of about 300 nodes, proposed the move from LSF to Slurm, and led the archive and transfer services through growth of 2 to 3PB a month and a sustained 60 to 75Gbps of transfer traffic.

- Currency: NVIDIA Academy AI infrastructure and networking courses, listed under credentials. Planned exam, November 2026: NVIDIA-Certified Professional, AI Networking.

Experience

Professional development, Apr 2026 to present

Since leaving EMBL-EBI in March 2026: Oxford Saïd Executive Finance Programme, May 2026; Chartered IT Professional status awarded by BCS, August 2026; Google Professional Cloud Architect, September 2026; NVIDIA Academy AI infrastructure and networking courses, September and October 2026. Available to start at short notice.

EMBL-EBI, Hinxton, UK. Sep 2016 to Mar 2026

Deputy Head of Operations, Jan 2021 to Mar 2026

- Accountable for reliability, cost and delivery of archive, transfer, identity and monitoring services. The estate held more than 4,500 Linux servers and VMs. More than 50,000 researchers used its public data, from over 80 countries. I worked with research leaders, archive owners, networking and security teams, most of them outside my reporting line.
- A three-stage reorganisation of Operations towards DevOps and SRE practices, which I designed in 2025. I reviewed the ITS risk register in 2024.
- About £520k secured through the SITSC infrastructure roadmap for a large CAD data transfer and processing project in 2025.
- A skills-based outsourcing model reduced specialist dependency and cut licensing costs by 25%.
- Advocated stretched VLANs and helped sequence the cutover with networking and service teams. EMBL-EBI completed its first data-centre migration with no service interruption. No addresses changed.
- Hired and mentored the IAM team. We moved EMBL's identity estate from OpenLDAP with bespoke schemas to Red Hat Identity Management with standard schemas and an Active Directory trust.
- Started and chaired the IAM change advisory group in 2023 and 2024. HR records became the authoritative source: no account without a signed contract, a single-use activation link with two-factor setup for joiners, and accounts closed on the contract end date.
- A member of the BioImage Archive Risk Management Board, as FIRE service owner, assessing infrastructure risk as that archive grew towards 100PB by 2030.
- All Transfer Services tickets now route to the Service Desk. I trained its staff ticket by ticket and wrote their documentation. Rundeck lets them run delegated actions without seeing scripts or credentials.

Compute, then archive, transfer and identity services lead, Sep 2016 to Dec 2020

- Joined in the compute team: about a year helping run three LSF farms of about 300 nodes, and from day one the 25-node LSF cluster that ran the transfer services as LSF-scheduled containers. I rebuilt it, with self-healing and monitoring scripts that adjusted LSF queue configuration, and proposed on cost grounds the move from LSF to Slurm that the institute made years later.
- FIRE took five archive teams, each with its own staging area, database and pull daemon, and replaced that with one HTTP upload API and S3-compatible reads. Old and new ran together. Reads moved first, then writes, each team switching when ready. Growth reached 2 to 3PB a month.
- I created and chaired the FIRE Users Group. Every team agreed to change the service contract from replicated in two data centres and two technologies to available in two data centres, eventually replicated to a second technology. It was cheaper and easier to maintain.
- Java and Spring Boot services on Kubernetes replaced the earlier Python stack, built through GitLab CI/CD with mandatory code review, unit and functional tests, and one image promoted unchanged through every environment. Provisioning in Puppet, Ansible and SaltStack.
- I wrote and ran the tender for an S3-compatible tape tier. Staged acceptance tests led to the PoINT Archival Gateway, deployed in August 2020; it writes S3 objects to IBM tape through a RAM cache.
- I created and chaired the Transfer Services Users Group. The final set of nine services is documented on one page. Quotas on two mounts gave capacity planning and quality of service.
- UK Biobank, 2020: 3.9PB ingested during the COVID-19 response; I proposed Globus over Aspera.
- Successive JANET/Jisc upgrades took the external link from 2x10Gbps to 40Gbps and then 100Gbps.
- I built cost models and TCO for AWS and GCP as active or backup storage options.

Bull SAS and Atos, Barcelona and Spain. Apr 2011 to Sep 2016

HPC systems engineer and trainer; technical lead, managed services in Barcelona; Technical Service Manager from 2015; Solutions Architect in the final period. Bull was acquired by Atos in 2014.

- Joined as an L3 HPC engineer and trainer for the Barcelona Supercomputing Center, CESCA and CSIC. I configured and operated Slurm and worked with LSF and the bullx suite.
- BSC MinoTauro, with NVIDIA Tesla M2090 GPUs, was my first full cluster installation: racks and water-cooled rear doors, InfiniBand and GPU cabling, CUDA drivers, GPU blade firmware and BIOS issues, Slurm with a highly available controller, acceptance node by node and then across the InfiniBand links. I trained the customer's administrators and maintained it afterwards.
- I designed the InfiniBand fabrics with Voltaire and Mellanox subnet management, and worked on storage from Lustre on DDN to Ceph up to 5PB and EMC Celerra up to 2PB.
- At BSC I moved a 1PB Lustre system to a new 2PB one, then added the old capacity to it, using bonded interfaces on the InfiniBand cards and Slurm-scheduled jobs.
- Acceptance with Linpack and IOzone; rack layout, cabling, power and cooling planning, commissioning. I contributed to tenders for national supercomputing procurements in Spain.
- Atos Tenerife HPC data centre: an underused cluster of about 700 nodes consolidated to 250, then rebuilt from bare metal with Katello, PXE and SaltStack in about 16 minutes a pass, with OpenStack and Hortonworks Hadoop on top. That proof of concept helped my business unit secure ClickFox service delivery, about EUR 1m in associated internal services revenue.
- Co-founded Bull Spain's managed-services group in Barcelona with its manager and an ITIL lead. I owned the technical design, technical client management, the L1 and L2 engineers' operating documentation and mentoring, and L3 escalation.
- Designed and deployed a SaltStack orchestration system for about 4,500 Unix and Windows servers, and set up FreeIPA to federate authentication for about 400 Active Directory help-desk operators.
- Automated OpenStack deployment for FIWARE's distributed infrastructure. Multi-year roadmaps for availability, disaster recovery and continuity across private cloud, big data and HPC.
- Migrated four clients' assets from a BMC tool to Atos ServiceNow, and contributed to ISO 20000 and ISO 27001 certification work.
- Bull's Top Contributor recognition in three consecutive years; Atos Global Talent Pool in 2016.

Barcelona Media, now Eurecat, Barcelona. May 2007 to Mar 2011

Research support coordinator and infrastructure lead

- Ran the IT and research platform function as the organisation grew from about 50 to 250 people: Linux system administration, identity, messaging, monitoring, backup and grid systems, with Puppet and SVN for version-controlled provisioning and Ganglia and Zabbix for monitoring.
- Hardware and software tenders, including Grid Engine for the research compute grid. An eight-GPU compute system bought from Bull was running on Debian with Sun Grid Engine within a week.

Professional credentials

Professional standing: Chartered IT Professional, CITP MBCS, BCS, awarded August 2026.

Cloud: Google Professional Cloud Architect, September 2026; Open Telekom Cloud Architect, April 2026.

AI and GPU infrastructure: NVIDIA Academy, September 2026: AI Factory Reference Architectures; AI Infrastructure and Operations Fundamentals; Introduction to Networking; InfiniBand Essentials; Cumulus Linux Essentials; NetQ Deployment and Installation; MLXLink and MLXCables; SONiC Essentials. NVIDIA Academy, October 2026: InfiniBand Network Administration. NVIDIA Deep Learning Institute, 2025: Fundamentals of Accelerated Data Science; Adding Knowledge to LLMs.

Service management: ITIL 4 Strategist Direct, Plan and Improve, 2026; PRINCE2 Agile Practitioner and Foundation, 2020; EXIN IT Service Management Foundation, ISO/IEC 20000, 2012.

Earlier technical: Red Hat RHCSA and RHCE, 2011, lapsed; Oracle Database 11g OCP, 2015.

Education

Cambridge Judge Business School, Chief Technology Officer Programme, 2025.

Oxford Saïd Business School, Executive Finance Programme, May 2026.

BSc Computer Science and Engineering, La Salle, Universitat Ramon Llull, Barcelona, 2005. Final-year project at the University of Strathclyde, 2006.